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Mock Observations for the CSST Mission: Multi-Channel Imager—The Cluster Field

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Abstract

The Multi-Channel Imager (MCI), one of the instruments aboard the Chinese Space Station Survey Telescope, is designed to simultaneously observe the sky in three filters, covering wavelengths from near-ultraviolet to near-infrared. With its large field of view ($7'.5 \times 7'.5$), MCI is particularly well-suited for observing galaxy clusters, providing a powerful tool for investigating galaxy evolution, dark matter and dark energy through gravitational lensing. Here we present a comprehensive simulation framework of a strong lensing cluster as observed by MCI, aiming to fully exploit its capabilities in capturing lensing features. The framework simulates a strong lensing cluster from the CosmoDC2 catalog, calculating the gravitational potential and performing ray-tracing to derive the true positions, shapes and light distribution of galaxies within the cluster field. Additionally, the simulation incorporates intra-cluster light (ICL) and spectral energy distributions (SEDs), enabling further strong lensing analyses, such as ICL separation from galaxy light and mass reconstruction combining strong and weak lensing measurements. This framework provides a critical benchmark for testing the MCI data pipeline and maximizing its potential in galaxy cluster research.

Key words: galaxies: clusters: general – Galaxy: general – gravitational lensing: strong – telescopes

1. Introduction

The Chinese Space Station Survey Telescope (CSST, Zhan 2011, 2021; Cao et al. 2018; Gong et al. 2019) is a space telescope orbiting the Chinese Space Station (CSS). Aiming at addressing fundamental questions in cosmology and astrophysics, such as the large-scale structure of the Universe and the mysteries of dark matter and dark energy, CSST will be equipped with several modules, including the main Survey Camera (SC), a Multi-Channel Imager (MCI), an Integrated Field Unit spectrometer (IFS), an exoplanet imaging coronagraph, and a Tera-hertz module.

Among these modules, the MCI offers simultaneous imaging across three filters (CBU: 0.255–0.43 μm, CBV: 0.43–0.7 μm, CBI: 0.7–1.0 μm) that cover near-ultraviolet (NUV), optical and near-infrared (NIR) wavelengths, and a large field-of-view (FoV) of $7'.5 \times 7'.5$ with a pixel size of $0''.05$. These design features make MCI particularly effective for capturing high-resolution, multi-wavelength images of strong lensing clusters, facilitating a variety of detailed studies with the observed multiple images, which provide valuable insights into the Universe, e.g., mapping the dark matter distribution within clusters (Massey et al. 2015; Jauzac et al. 2019; Meneghetti et al. 2020, 2023; Bergamini et al. 2023; Kawai & Meneghetti 2024), investigating the nature and behavior of dark energy (Jullo et al. 2010; Magaña et al. 2018; Caminha et al. 2021), studying the formation and evolution of galaxies (Welch et al. 2023; Asada et al. 2023; Mowla et al. 2024), and exploring the expansion rate of the Universe (Suyu et al. 2017; Courbin et al. 2018; Grillo et al. 2018; Millon et al. 2020; Acebron et al. 2023; Pascale et al. 2024).

Interpreting strong lensing features requires careful lens modeling based on a comprehensive understanding of astrophysical and instrumental systematic errors (Acebron et al. 2017; Chirivì et al. 2018; Granata et al. 2022). In this work, we create a simulated cluster field and employ a ray-tracing algorithm (Li et al. 2016) to model the lensing effects for all line-of-sight galaxies within the field. This simulation serves as a testbed for validating the MCI data pipeline by incorporating detailed instrumental effects, which is discussed in a companion paper (Yan et al. 2026), and for evaluating the performance of various lens modeling algorithms, e.g., LENSTOOL (Jullo et al. 2007), CURLING (Xie et al. 2024), while also identifying the impact of other sources of systematics. As the contamination from intra-cluster light (ICL, Zwicky 1937; DeMaio et al. 2015; Montes 2019), which consists of diffuse light emitted by stars unbound to any individual cluster galaxy, poses a significant challenge in extracting the light of lensed arcs and member galaxies (Montes & Trujillo 2014; Di Criscienzo et al. 2017)—both critical components for accurate lens modeling, we include ICL in our simulation to enable us to develop strategies for disentangling these light components in real observations. Furthermore, MCI's large FoV allows observations of cluster outskirts extending to several Mpc, providing valuable weak lensing information. By combining weakly and strongly lensed galaxies, we can achieve a more comprehensive mass reconstruction of the cluster (Quinn Finney et al. 2018; Strait et al. 2018; Niemiec et al. 2023; Patel et al. 2024). Since photometry is essential for both measurements—particularly for weak lensing—we incorporate spectral energy distributions (SEDs) for all galaxies, ensuring that photometric measurements are available for further weak lensing analyses with the simulated cluster.

The work is structured as follows. In Section 2 we introduce the galaxy data from the CosmoDC2 catalog, which serves as the foundation for our simulation framework. In Section 3, we introduce the methodology for constructing mock images of the simulated cluster within the framework, outlining the steps involved in incorporating gravitational lensing effects. Section 4 expands the simulation framework by introducing additional components, such as the cluster ICL, the SEDs of the galaxies, and simplified instrumental effects. Finally, Section 5 provides a summary and conclusion. Throughout the paper, we assume a flat wCDM cosmology ($\Omega_{\Lambda,0} = 0.7$, $\Omega_m = 0.3$, $h = 0.7$) with a constant dark energy equation of state parameter $w = w_0 = -1$.

2. Cluster Catalog and Simulation Setup

In this work, the simulated cluster field for MCI is constructed based on the CosmoDC2 (Korytov et al. 2019) catalog, which provides information about the cluster member galaxies and the line-of-sight galaxies.

2.1. Cosmological Simulation

The CosmoDC2 catalog is created from the "Outer Rim," a large volume cosmological N-body simulation with a box size of 4.225 Gpc and $(10{,}240)^3$ tracer particles, achieving a mass resolution of $m_p = 2.6 \times 10^9 M_\odot$.

Halo catalogs are generated using a parallel Friends-of-Friends (FoF) halo finder with a linking length of $b = 0.168$ and a minimum requirement of 20 particles per halo. These halos are then linked across snapshots using a particle-membership algorithm (Rangel et al. 2020) to construct the halo merger tree, which serves as the foundation for the construction of halo lightcones. Galaxies are placed within these halo lightcones by first resampling from the galaxy catalog produced by the UNIVERSEMACHINE (Behroozi et al. 2019) code and refining them through empirical modeling to ensure realistic property distributions. Each galaxy is then matched with a counterpart in the galaxy library generated with the GALACTICUS (Benson 2012) code to incorporate a comprehensive set of physical properties.

The final catalog contains ~2.26 billion galaxies in a 440 deg^2 field that spans $0 < z < 3$. Each galaxy has 551 listed properties including stellar mass, fluxes in LSST and SDSS bands, shape information, etc.

2.2. Galaxies in the CosmoDC2 Catalog

2.2.1. Cluster Member Galaxies

The MCI cluster simulation is performed by extracting the cluster galaxies from the CosmoDC2 catalog. We focus on a cluster with a virial mass of $M_{\rm vir} = 1.13 \times 10^{15} M_\odot$ at redshift of $z_L = 0.3$, which stands for a typical cluster for strong gravitational lensing studies. The cluster field is cropped to a region of $7\farcm5 \times 7\farcm5$ to align with the MCI observational setup. Within this field, we identify a total of 211 cluster member galaxies. Detailed galaxy properties from the catalog include positions, morphological parameters (semimajor and semiminor axes, position angle, Sérsic profile index), and magnitudes in the LSST g, r, and i bands, which represent the corresponding MCI CBU, CBV, and CBI bands. The morphological and flux parameters are further decomposed for disk, bulge, and total components, with each galaxy also assigned a bulge-to-disk ratio. These data are used to compute the gravitational potential of the cluster for ray tracing and to generate the MCI-observed image of the cluster field in the subsequent analyses.

2.2.2. Catalog of Source Galaxies

Within the $7\farcm5 \times 7\farcm5$ lightcone of the simulated cluster, a total of 80,532 source galaxies are identified. Each of these galaxies (we classify them as "line-of-sight galaxies" in the following analyses) is characterized by the same set of properties as the cluster member galaxies described in Section 2.2.1. In addition, the redshift of each source galaxy is recorded to facilitate the computation of its light distribution after being gravitationally lensed by the cluster potential. The

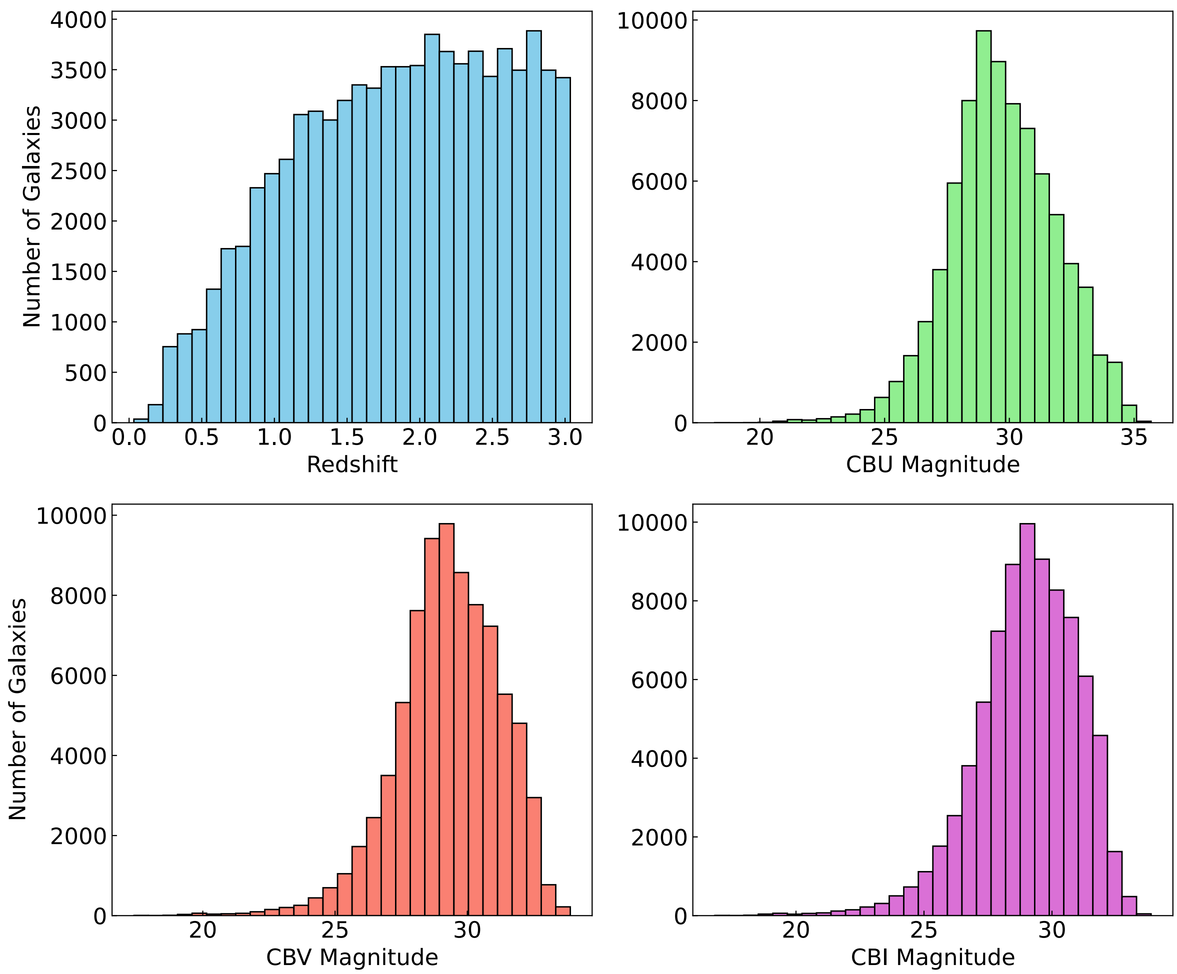

Figure 1. Redshift and CBU, CBV, CBI magnitude distributions of the 80,532 source galaxies within the 7′.5 × 7′.5 lightcone of the simulated MCI cluster.

source galaxies span a redshift range from $z_s = 0$ to $z_s = 3$. To illustrate their overall distribution, Figure 1 presents the redshift and magnitude distributions across the MCI CBU, CBV, and CBI bands for all line-of-sight galaxies.

3. Lensing Simulations Framework

3.1. Lensing Theory

Gravitational lensing (Kneib & Natarajan 2011; Meneghetti 2022) is a powerful astrophysical phenomenon that occurs when the light from a distant source, such as a galaxy or a quasar, is deflected by the gravitational field of a massive object, such as a galaxy cluster, lying between the source and the observer. This effect, predicted by Einstein's General Theory of Relativity, can magnify, distort the source, and create multiple images of the background sources.

To accurately account for the lensing effect, we perform ray-tracing, a method that traces the path of light rays as they travel through the gravitational field of the cluster. The process involves solving the lens equation, which relates the positions of the source β and the image θ,

$$\vec{\beta} = \vec{\theta} - \vec{\alpha}(\vec{\theta}), \tag{1}$$

where the deflection angle $\vec{\alpha}(\vec{\theta})$ is determined by the Newtonian gravitational potential $\Phi(\vec{\theta})$ and the geometry of the Universe, specifically the angular diameter distances between the lens L, source S, and observer O (D_{LS}, D_{OL}, and D_{OS}),

$$\vec{\alpha}(\vec{\theta}) = \frac{2}{c^2}\frac{D_{LS}}{D_{OL}D_{OS}}\nabla\Phi(\vec{\theta}). \tag{2}$$

The lens equation is typically highly nonlinear and can have multiple solutions, corresponding to the cases of multiple images as a result of strong gravitational lensing.

In addition to multiple images, another important feature of lensing distortion is the magnification of the sources. Mapping the coordinates from the source plane to the image plane involves the *Jacobian Matrix*,

$$A \equiv \frac{\partial \vec{\beta}}{\partial \vec{\theta}} = \left(\delta_{ij} - \frac{\partial \alpha_i(\vec{\theta})}{\partial \theta_j}\right) = \left(\delta_{ij} - \frac{\partial^2 \Psi(\vec{\theta})}{\partial \theta_i \partial_j}\right) = \begin{pmatrix} 1-\kappa-\gamma_1 & -\gamma_2 \\ -\gamma_2 & 1-\kappa+\gamma_1 \end{pmatrix}, \tag{3}$$

where the lensing potential $\Psi(\boldsymbol{\theta}) = \frac{2}{c^2}\frac{D_{LS}}{D_{OL}D_{OS}}\int \Phi(D_l\boldsymbol{\theta}, z)\, dz$, is derived from the gravitational potential Φ by integrating along the line of sight, the convergence term $\kappa = \frac{1}{2}\Delta\Psi = \frac{1}{2}(\Psi_{11} + \Psi_{22})$ defines the isotropic transformation of images, and the shear terms $\gamma_1 = \frac{1}{2}(\Psi_{11} - \Psi_{22})$ and $\gamma_2 = \Psi_{12} = \Psi_{21}$, with the complex shear $\gamma = \gamma_1 + i\gamma_2$, delineate the stretching of images. The magnification is given by the inverse of the determinant of the Jacobian matrix,

$$\mu \equiv \det M = \frac{1}{\det A} = \frac{1}{(1-\kappa)^2 - |\gamma|^2}, \tag{4}$$

3.2. Cluster Potential

According to Equations (1) and (2), the solution of the lens equation for ray tracing is based on computing the lensing potential. In this simulation, the cluster is a combination of several components, each contributing to the overall gravitational potential:

1. *Dark matter halo*. The dominant component of the cluster's mass distribution is the dark matter halo, which is associated with the brightest cluster galaxy (BCG). We model this halo using the Navarro–Frenk–White (NFW) profile (Navarro et al. 1997),

$$\rho(r) = \frac{\rho_s}{\frac{r}{r_s}\left(1 + \frac{r}{r_s}\right)^2}, \tag{5}$$

   where ρ_s is the characteristic density of the profile, $r_s = r_{200}/c$ is the characteristic radius, with r_{200} defined as the radius within which the average density of the cluster is 200 times the critical density of the Universe (ρ_c) at the cluster redshift, c defined as the concentration parameter. As provided in the catalog, the halo has a virial mass of $M_{\rm vir} = 1.13 \times 10^{15} M_\odot$, and a concentration of $c = 3.86$.
2. *Subhalos*. We employ the Pseudo-Isothermal Elliptical Mass Distribution (PIEMD, Elíasdóttir et al. 2007) to model the subhalos associated with member galaxies. The PIEMD potential is defined as,

$$\rho(r) = \frac{\rho_0}{\left(1 + \frac{r^2}{r_{\rm core}^2}\right)\left(1 + \frac{r^2}{r_{\rm cut}^2}\right)}, \tag{6}$$

   where ρ_0 denotes the central density of the profile, while $r_{\rm core}$ and $r_{\rm cut}$ are the core radius and the truncation radius, respectively. Within the regions where $r_{\rm core} < r < r_{\rm cut}$, the profile behaves as $\rho \sim r^{-2}$, while it transitions to $\rho \sim r^{-4}$ in the outer regions where $r > r_{\rm cut}$. Given that PIEMD parameters, i.e., ρ_0, $r_{\rm core}$, and $r_{\rm cut}$, are not provided in the original CosmoDC2 catalog, we assign these values to each galaxy based on its luminosity, using the scaling relations (see Equation 4 in Jullo et al. 2007) as used in strong lensing modeling:

$$\begin{cases} \sigma_i^{\rm gal} = \sigma^{\rm ref} 10^{0.4\frac{m^{\rm ref} - m_i}{\alpha}} \\ r_{{\rm core},i}^{\rm gal} = r_{\rm core}^{\rm ref}\, 10^{0.4\frac{m^{\rm ref} - m_i}{2}} \\ r_{{\rm cut},i}^{\rm gal} = r_{\rm cut}^{\rm ref}\, 10^{0.4\frac{2(m^{\rm ref} - m_i)}{\beta}}, \end{cases} \tag{7}$$

   The typical core radius, cut radius and velocity dispersion for the scaling relations are set to $r_{\rm core}^{\rm ref} = 1.0$ kpc, $r_{\rm cut}^{\rm ref} = 12.0$ kpc, $\sigma^{\rm ref} = 200.0\ {\rm km\ s^{-1}}$ in this work.

The deflection angle map, calculated as the gradient of the lensing potential (Equation (2)), is then used to derive the image distortions for each source in the catalog (Equation (1)). In Figure 2, we show the convergence map of the simulated cluster and the map of light rays from the lens plane to the source plane, illustrating the effects of gravitational lensing.

3.3. Ray Tracing for Line-of-sight Galaxies

To generate the mock image, we consider the light distribution of every galaxy within the field. Each galaxy is modeled as a combination of a bulge and a disk, both described by a Sérsic profile:

$$I(r) = I_e \exp\left\{-b_n\left[\left(\frac{r}{r_e}\right)^{1/n} - 1\right]\right\}, \tag{8}$$

where $I(r)$ is the intensity at radius r, I_e is the central intensity, r_e is the effective radius, n is the Sérsic profile index, and b_n is a constant parameter. As detailed in Section 2.2.2, the Sérsic profile parameters, along with the galaxy's position, shape, and magnitude, are provided by the CosmoDC2 catalog.

For galaxies lying behind the cluster, we calculate the observed source position θ from its original position β based on the deflection angle α, accounting for the gravitational lensing effect. Here, we assume the lens as the single-plane cluster, as the primary goal is to compare lens modeling codes and support companion work that focuses on MCI instrumental simulation (Yan et al. 2026). More complex scenarios involving multi-lens planes will be explored in future studies.

Once the light distributions of all galaxies have been computed, we superimpose them to generate the complete

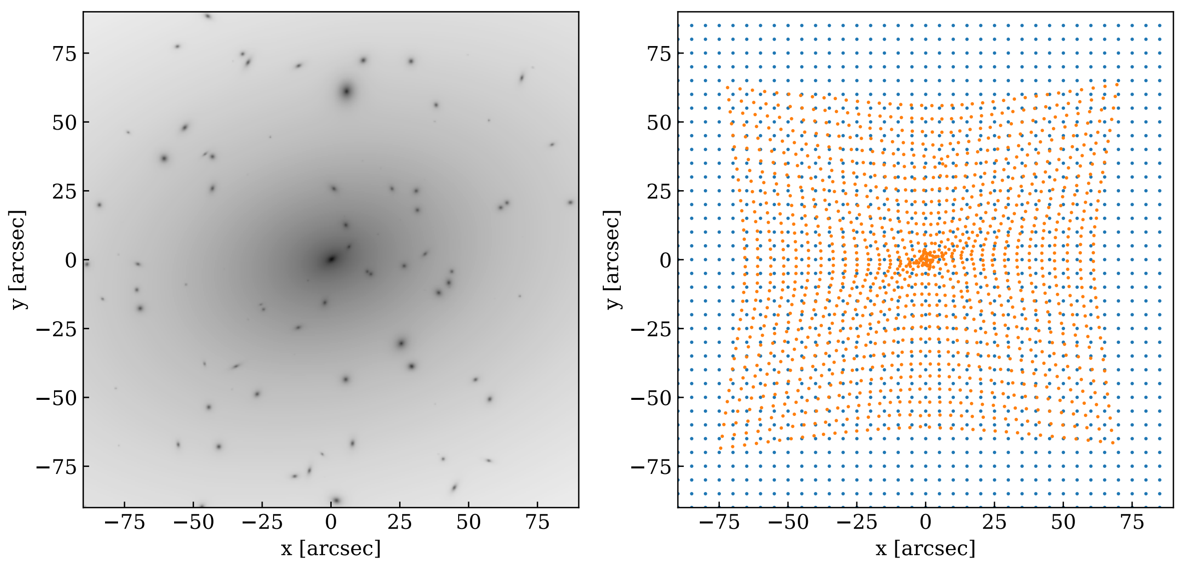

Figure 2. Left panel: Convergence map of the simulated MCI cluster. Right panel: Illustration of the light rays on the lens plane (blue dots) and their mapped positions on the source plane (orange dots) due to the deflection of gravitational lensing.

simulated image of the cluster field. In Figure 3, we present the generated image in the MCI CBV band. For clarity, only the central $60'' \times 60''$ is displayed, highlighting the lensing features.

In addition to ray tracing the galaxies provided by the CosmoDC2 catalog, we further add 20 galaxies to incorporate strongly lensed arcs, which will be valuable for strong lensing analyses with this simulated cluster. These galaxies are randomly extracted from the same catalog, while we adjust their positions to guarantee the formation of giant arcs. An example of a source galaxy and its corresponding lensed arc is shown in Figure 4. Note that in real observations, the source and the lensed image cannot be seen simultaneously. Here, they are presented together for clarity in illustrating positional and morphological changes.

3.4. Computational Accelerating with JAX

Given that the catalog within the MCI FoV contains over 80,000 sources, and the lensing potential is computed across a grid of 9000×9000 pixels, the process of generating the simulated images originally requires over 10 hours, which is highly time consuming. To address this, we optimize the simulation algorithm based on the JAX framework (Bradbury et al. 2018). This enhancement allows us to perform ray-tracing for an entire simulated cluster system in only 6 minutes, achieving a two-order-of-magnitude speedup compared to the previous implementation.

4. Simulation Enhancements

To prepare for MCI observations of galaxy clusters, we simulate a cluster with strong lensing effect, as introduced in

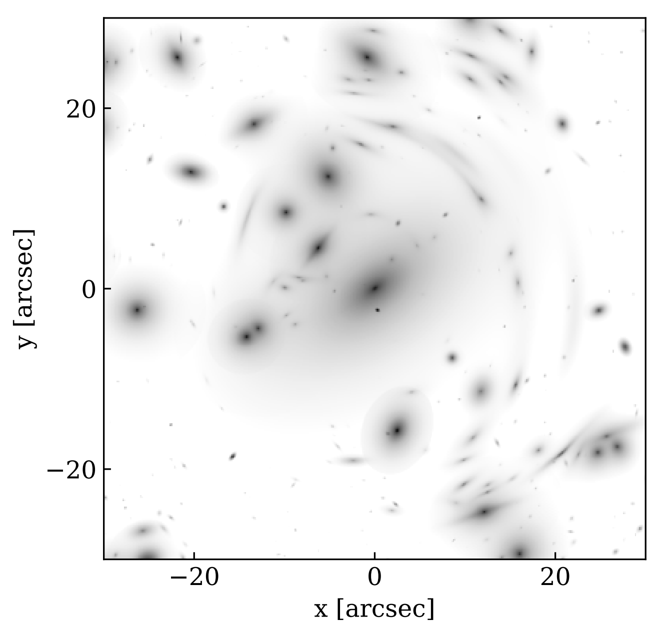

Figure 3. Galaxy distributions within the central region ($60'' \times 60''$) of the simulated cluster, accounting for the gravitational lensing effect.

the previous sections. This simulated cluster serves as an essential testbed for the MCI team to implement realistic instrumental effects, such as point-spread function (PSF) convolution, charge transfer inefficiency (CTI), nonlinearity, and other detector effects, which are detailed in the companion paper by Yan et al. (2026). Additionally, simulated multiple-image families, coupled with the known cluster lens potential, provide a valuable resource for testing and comparing the performance of various lens modeling codes, which can help with the development of modeling algorithms in the strong

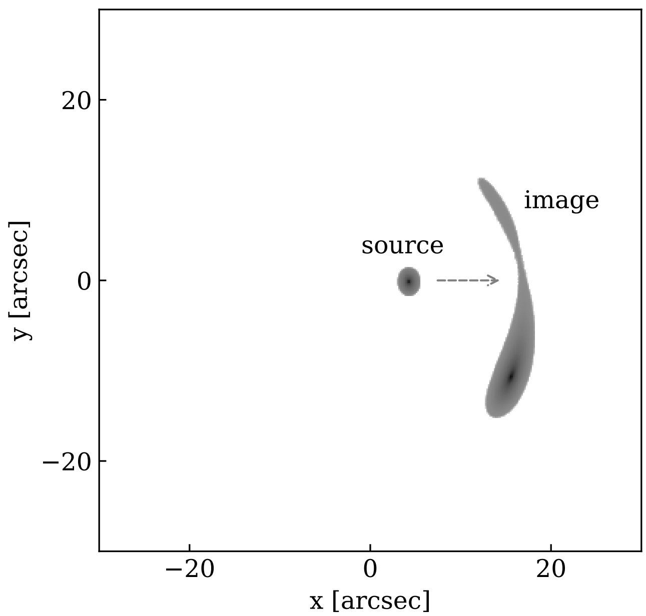

Figure 4. An illustration of the multiply lensed arc. The "source" shows where the source is originally lying. The "image" indicates its appearance and position after being deflected by the lens.

lensing community. Beyond the primary simulated cluster for scientific validation, we also incorporate several additional simulations to facilitate further testing with the cluster in the context of MCI. These additional simulations cover aspects including the ICL of the cluster, and the SEDs for all member galaxies and line-of-sight galaxies, and the implementation of instrumental effects.

4.1. Simulation of Intra-cluster Light

ICL is the diffuse and faint stellar population that spreads through clusters of galaxies, stripped from their host galaxies. First observed by Zwicky in 1937 (Zwicky 1937) and later confirmed in the Coma Cluster (Zwicky 1951, 1957), ICL has become an essential feature in the study of galaxy clusters. Over the last two decades, numerous studies have shown that the ICL is closely related to the nature of galaxy clusters, in particular their dynamical evolution history (Contini et al. 2014, 2018, 2023; DeMaio et al. 2015, 2018; Montes 2019; Chen et al. 2022). The light of ICL is often entangled with the light from galaxies, as well as with lensed arcs (Montes & Trujillo 2014; Di Criscienzo et al. 2017), complicating the extraction of these features for strong lensing analyses. Moreover, the ICL can also affect the estimation of galaxy shapes and counts in weak lensing regime (Gruen et al. 2019). Facing this challenge, we incorporate the ICL into our simulated cluster field, aiming to refine algorithms for accurately subtracting its contribution.

To simulate the ICL, we first model both the mass and luminosity distributions of this diffuse stellar population. The mass distribution of the ICL is assumed to follow the NFW profile (Zibetti et al. 2005; Pillepich et al. 2018; Zhang et al. 2019), with the parameters $M_{\rm ICL}$ and $c_{\rm ICL}$. The total mass of the ICL is estimated using an empirical relation from Equation (5) in Purcell et al. (2007), which yields the ICL fraction $f_{\rm ICL} = M_*^{\rm diffuse}/M_*^{\rm total}$, with $M_*^{\rm total}$ defined as the total stellar mass of the cluster, and $M_*^{\rm diffuse}$ defined as the mass of the ICL. We calculate $f_{\rm ICL} = 0.2162$, $M_*^{\rm total} = 2.16 \times 10^{13} M_\odot$, $M_*^{\rm diffuse} = 4.66 \times 10^{12} M_\odot$, and make a simplified assumption that the ICL follows the concentration of the dark matter halo, with $c = 3.86$, although studies suggest that the ICL is typically more concentrated than the total mass distribution (Cooper et al. 2013; Lowing et al. 2015; Rodriguez-Gomez et al. 2016; Wang et al. 2021). After constructing the mass distribution, we simulate the luminosity distribution of the ICL, for the stellar mass-light ratio, we follow the coefficient in the modified relation of Equation 2 in Chen et al. (2022), which approximately equals to unity. We then account for the effects of cosmic dimming and *k*-corrections (Hogg 1999),

$$\mu = \mu_0 \left(\frac{M_h}{M_{h,0}}\right)^{-1/3} \times \left(\frac{D_a}{D_{a,0}}\right)^2 \times \left(\frac{D_{l,0}}{D_l}\right)^2 \times 10^{-0.4(m-m_0)} \quad (9)$$

where μ_0 is the surface brightness profile of a cluster at redshift z_0 ($D_{a,0}$ and $D_{l,0}$ are the corresponding angular diameter distance and luminosity distance, respectively), with mass $M_{h,0}$ and rest-frame magnitude m_0. The term $\left(\frac{D_a}{D_{a,0}}\right)^2 \times \left(\frac{D_{l,0}}{D_l}\right)^2$ accounts for the cosmic dimming effect, while $10^{-0.4(m-m_0)}$ represents the *k*-correction term.

Initially, the ICL distribution is assumed to be symmetric. To simulate a more realistic scenario, we introduce asymmetries into the ICL distribution based on the spatial distribution of cluster galaxies. This is achieved by computing the second-order moment of the spatial distribution of the member galaxies and generating a normalized two-dimensional Gaussian function with these parameters. This Gaussian is then convolved with the symmetric ICL light distribution, incorporating the alignment with the observed dispersion of cluster galaxies.

Figure 5 illustrates the ICL simulation, showing the symmetric and asymmetric light distributions (upper right and lower left panels), along with the superimposed ICL and galaxy distributions (lower right panel).

4.2. Spectral Energy Distributions for Line-of-sight Galaxies

The MCI's notable advantages over other space-based observations include its designed depth up to 30 AB magnitude in optical and NUV bands, along with the large

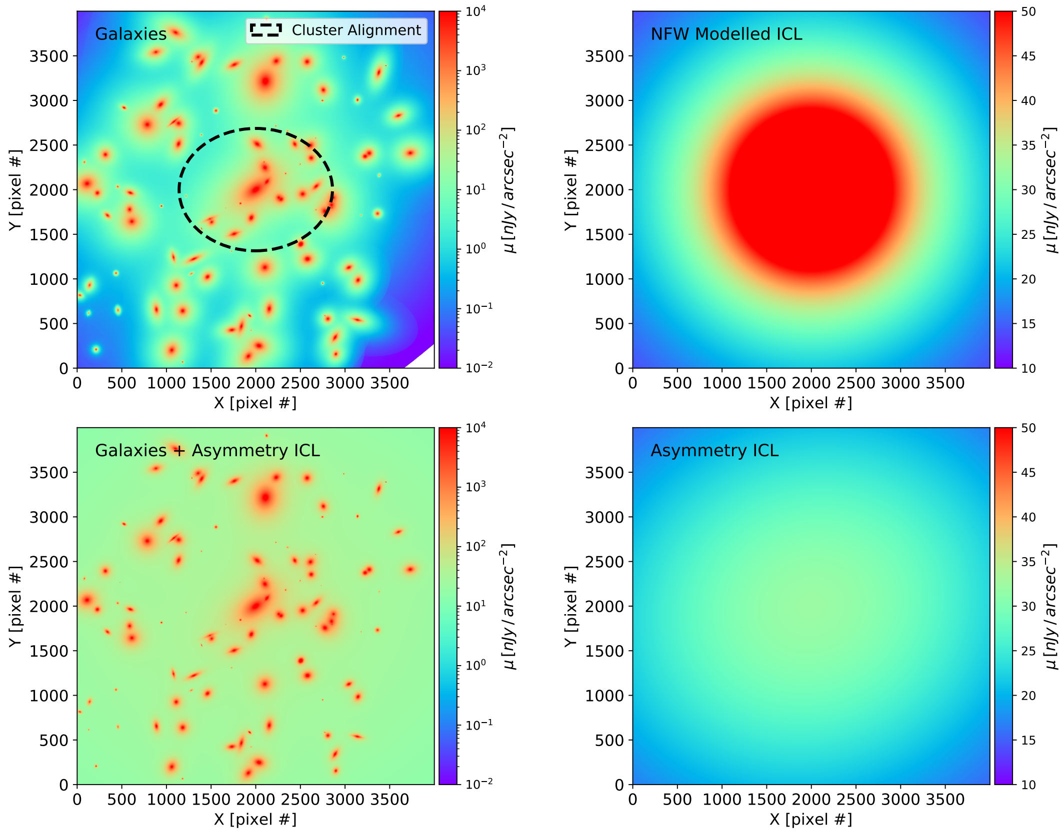

Figure 5. Illustration of the ICL mock in the CBV band. The image is presented within 4000 × 4000 pixels (corresponding to $200'' \times 200''$). Top Left: Luminosity distribution of cluster member galaxies. The second-order moment of the galaxy position distribution is represented by the black dashed ellipse in the center of the figure. Top Right: Simulated ICL image modeled using the NFW profile. Bottom Left: Mock of asymmetric ICL luminosity distribution. Bottom Right: The luminosity distribution of the sum of the asymmetric ICL and the member galaxies of the cluster.

FoV of $7'.5 \times 7'.5$. These characteristics enable a comprehensive mass map of galaxy clusters, extending several Mpc, by complementing weak lensing measurements with strong lensing observations. Consequently, accurate photometry for the large amount of weakly lensed sources is necessary to interpret their properties. Here, we present the simulation of SEDs for all the line-of-sight galaxies within the cluster field, to take full advantage of MCI's multi-band photometric capabilities, particularly its narrow-band sensitivity. Since the galaxies provided in the original CosmoDC2 catalog are generated by combining the UNIVERSEMACHINE and GALACTICUS models, and their star formation histories (SFHs) are not necessarily consistent or complete, we generate the SEDs based on the photometries.

We begin by constructing SED templates based on single stellar population (SSP) spectra from (Maraston 2005). The SSP templates span wavelengths from 100 to 10,000 Å, with stellar population ages ranging from 10^3 yr to 15 Gyr, and metallicities from [Z/H] = −2.25 to [Z/H] = 0.67. In this study, only four values of metallicity, [Z/H] = (−1.35, −0.33, 0, 0.35), are considered, which are sufficient to simulate the SEDs for the vast majority of galaxies. For a given metallicity, we stack the SSP into a composite stellar population (CSP) spectrum according to an exponentially declining star formation history,

$$\mathrm{SFR}(t) \sim e^{-(t-t_0)/\tau} \tag{10}$$

where τ is fixed at 1.0 Gyr, and t_0 is set as eight typical values from 10^7 yr to 12 Gyr. The fluxes in the CSP are set to zero for rest wavelengths shorter than 912 Å to model the absorption of ultraviolet photons by cold gas within galaxies. This process results in 32 continuous spectra of the stellar population, covering four metallicities and eight stellar ages.

Next, we used Gaussian functions to construct the emission line templates for ionized gas within galaxies. The line widths of the emission lines are uniformly taken to correspond to a velocity dispersion of 250 km s^{-1}, whereas the flux ratios of the emission lines are determined by the metallicity of the galaxies (for details, refer to Feng et al. in preparation). To match the stellar continuum templates, the ionized gas templates also include four metallicities. The final SED template is constructed by superimposing the emission line spectrum with the corresponding metallicity on the stellar

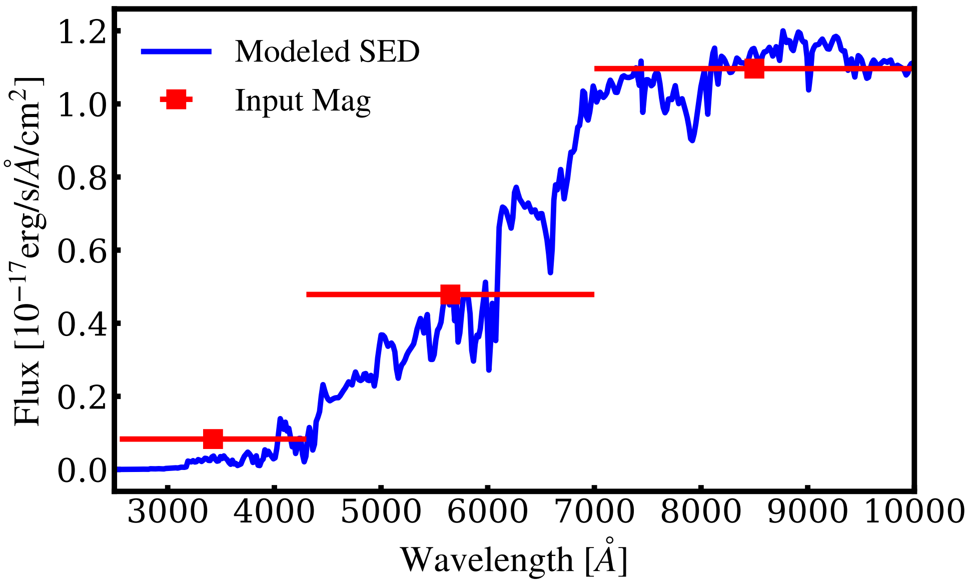

Figure 6. An example of SED mocking. The three red squares represent the input magnitudes ($m_{\rm CBU}$ = 24.1, $m_{\rm CUV}$ = 22.2, and $m_{\rm CBI}$ = 21.3) of a galaxy with a redshift of z = 0.53. The error bars represent the wavelength coverage of three bands. The blue line is the modeled spectrum of this galaxy.

population continuum spectrum. During the superimposition, the intensity of the emission line spectrum is determined by the equivalent width of Hα, EW(Hα), which can be approximated by (Fumagalli et al. 2012; Mármol-Queraltó et al. 2016):

$$\mathrm{EW(H}\alpha)/\text{Å} = 100(t_0/\,\mathrm{Gyr})^{-1.5}, \tag{11}$$

where t_0 is the stellar population age of the galaxy.

Using the SED templates, we simulate the SED of a galaxy in the wavelength range from 2500 to 10,000 Å according to the apparent magnitudes in the three MCI bands and the redshift of the galaxy. The overall shape of the SED is first simulated. For a galaxy with a redshift of z, we need to shift all 32 SED templates at the rest wavelength to the redshift z and calculate the magnitudes of the redshifted SED templates in the three bands of CBU, CBV, and CBI. The best SED template is selected which minimizes the difference between the galaxy and SED template magnitudes. Second, we consider the change in the shape of the SED caused by dust attenuation. We compare the CBU − CBI color of the best template with that of the input catalog. If the CBU − CBI color in the input catalog is redder, we redden the best template according to the extinction law of Calzetti et al. (1994) to obtain the observed SED shape of the target source. If not, the best template will be adopted as the observed SED shape. Finally, we calibrate the absolute flux of the best template using the magnitude in the CBV band. In Figure 6, we display one example of the SED simulation. The three red squares represent the apparent magnitudes in the MCI bands provided by the input catalog, the blue spectrum shows the simulated complete SED of the galaxy, demonstrating the fit between the data and the simulated SED template.

4.3. Instrumental Effects

Although a detailed treatment of instrumental effects is presented in the companion paper, here we provide a simplified version of incorporating the PSF and noise into the simulated image. As detailed in Yan et al. (2026), the Nyquist-sampled PSF for MCI is calculated by:

$$\mathrm{PSF} = |\mathrm{FFT}(P)|^2, \tag{12}$$

where P is the complex pupil function $P = Ae^{i\cdot 2\pi \mathrm{OPD}/\lambda}$, with the aperture function A, and the optical path difference (OPD) function at wavelength λ. This simulation for MCI PSF generates a set of 100 PSFs across the entire FoV. For simplicity, we convolve the simulated image with one of these PSFs.

The noise (Cao et al. 2022) to be added is calculated as:

$$\sigma = \frac{\sqrt{(S_{\rm img} + S_{\rm sky}) \times t_{\rm expo} + N_{\rm expo} \times R^2}}{t_{\rm expo}}, \tag{13}$$

where $S_{\rm img}$ and $S_{\rm sky}$ are the fluxes of the sources and sky background, $N_{\rm expo}$ and $t_{\rm expo}$ are the number of exposures, and total exposure time, respectively, R is the readout noise. For MCI observations, $S_{\rm sky}$ is calculated with the sky surface brightness $20.48\ \mathrm{mag\ arcsec^{-2}}$, $N_{\rm expo} = 2$, $t_{\rm expo} = 300$s, $R = 5e^-$.

By convolving the simulated image with the PSF and adding noise, we generate a realistic representation of the observed field (Figure 7). This processed image can be used by the strong lensing community to further test and refine their lens modeling codes, providing a robust tool for algorithm development and performance evaluation.

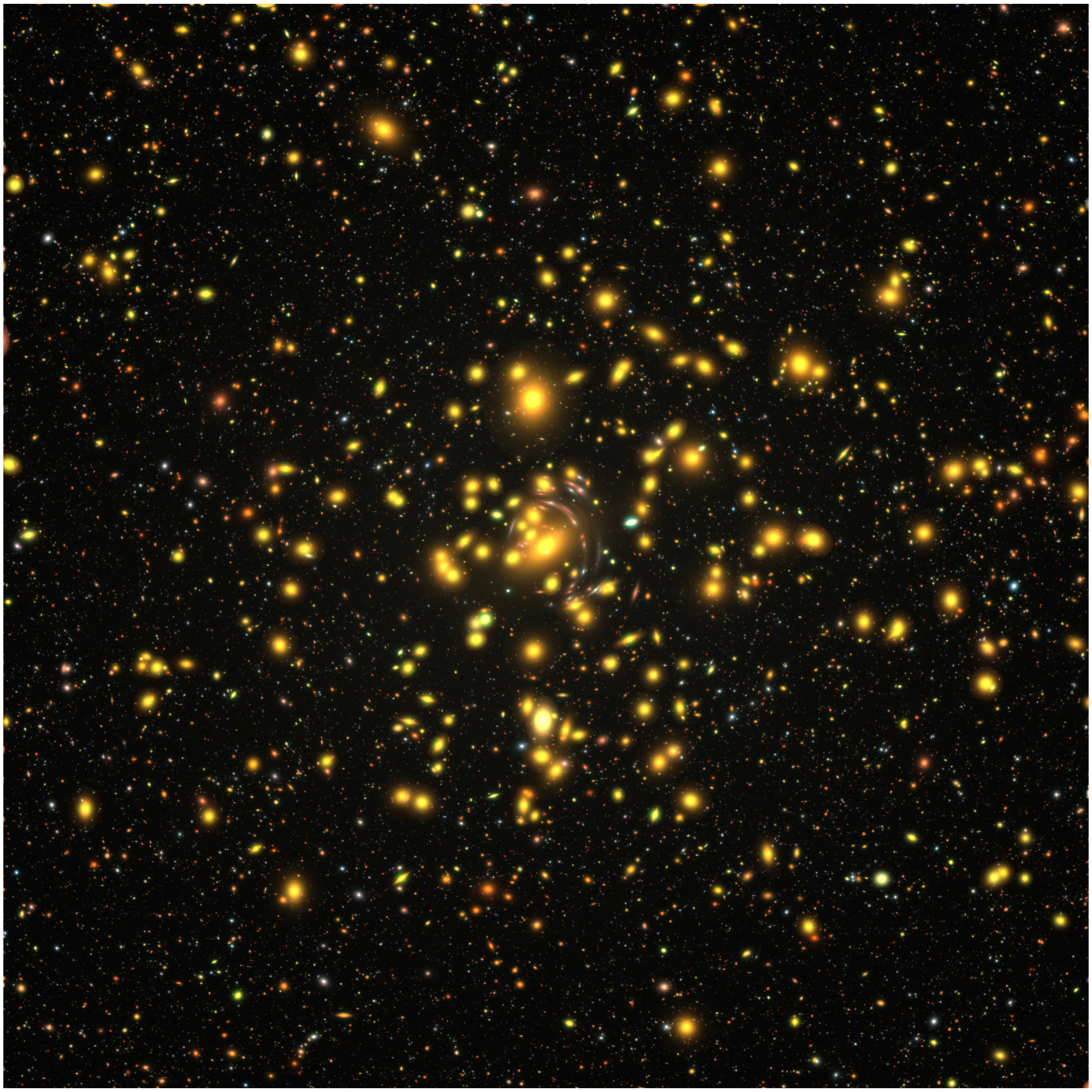

Figure 7. Superimposition of the galaxy distributions and the ICL within the full MCI FoV, presented as a composite image combining the CBU, CBV, and CBI bands. Instrumental effects, including the PSF and noise, are incorporated.

5. Summary

We present a comprehensive simulation framework for a galaxy cluster with gravitational lensing effect for the upcoming CSST-MCI, incorporating ray-tracing for all galaxies along the line-of-sight within a wide FoV of $7\overset{'}{.}5 \times 7\overset{'}{.}5$. Building upon the CosmoDC2 galaxy catalog, we construct a cluster lens model consisting of main dark matter halo and subhalos corresponding to member galaxies, and we calculate the light distributions of all galaxies while accounting for lensing deflections. This simulation provides a realistic and detailed representation of a galaxy cluster deep field as observed by CSST-MCI, with primary data products including mock images in the MCI CBU, CBV, and CBI bands, the catalog of the lensing potential, and a catalog of the simulated multiple image families. The simulation framework also integrates ICL, and SEDs of galaxies, and instrumental effects, enhancing its applicability for further analysis. Beyond improving our understanding of the physical processes within galaxy clusters, this simulation framework serves as a crucial tool for integrating instrumental effects into the data processing pipelines. Moreover, it establishes a strong foundation for advancing cluster lens modeling and will be instrumental in future cosmological studies, particularly those related to dark matter, galaxy evolution, and large-scale structure formation using CSST-MCI data.

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References

Acebron, A., Jullo, E., Limousin, M., et al. 2017, MNRAS, 470, 1809
Acebron, A., Schuldt, S., Grillo, C., et al. 2023, A&A, 680, L9
Asada, Y., Sawicki, M., Desprez, G., et al. 2023, MNRAS, 523, L40
Behroozi, P., Wechsler, R. H., Hearin, A. P., & Conroy, C. 2019, MNRAS, 488, 3143
Benson, A. J. 2012, NewA, 17, 175
Bergamini, P., Acebron, A., Grillo, C., et al. 2023, ApJ, 952, 84
Bradbury, J., Frostig, R., Hawkins, P., et al., 2018 JAX, : Composable transformations of Python+NumPy programs
Calzetti, D., Kinney, A. L., & Storchi-Bergmann, T. 1994, ApJ, 429, 582
Caminha, G. B., Suyu, S. H., Grillo, C., & Rosati, P. 2021, A&A, 657, A83
Cao, Y., Gong, Y., Meng, X.-M., et al. 2018, MNRAS, 480, 2178
Cao, Y., Gong, Y., Zheng, Z.-Y., & Xu, C. 2022, RAA, 22, 025019
Chen, X., Zu, Y., Shao, Z., & Shan, H. 2022, MNRAS, 514, 2692
Chirivì, G., Suyu, S. H., Grillo, C., et al. 2018, A&A, 614, A8
Contini, E., De Lucia, G., Villalobos, Á., & Borgani, S. 2014, MNRAS, 437, 3787
Contini, E., Jeon, S., Rhee, J., Han, S., & Yi, S. K. 2023, ApJ, 958, 72C
Contini, E., Yi, S. K., & Kang, X. 2018, MNRAS, 479, 932
Cooper, A. P., D'Souza, R., Kauffmann, G., et al. 2013, MNRAS, 434, 3348
Courbin, F., Bonvin, V., Buckley-Geer, E., et al. 2018, A&A, 609, A71
DeMaio, T., Gonzalez, A. H., Zabludoff, A., Zaritsky, D., & Bradač, M. 2015, MNRAS, 448, 1162
DeMaio, T., Gonzalez, A. H., Zabludoff, A., et al. 2018, MNRAS, 474, 3009
Di Criscienzo, M., Merlin, E., Castellano, M., et al. 2017, A&A, 607, A30
Elíasdóttir, Á., Limousin, M., Richard, J., et al. 2007, arXiv:0710.5636
Fumagalli, M., Patel, S. G., Franx, M., et al. 2012, ApJL, 757, L22
Gong, Y., Liu, X., Cao, Y., et al. 2019, ApJ, 883, 203
Granata, G., Mercurio, A., Grillo, C., et al. 2022, A&A, 659, A24
Grillo, C., Rosati, P., Suyu, S. H., et al. 2018, ApJ, 860, 94
Gruen, D., Zhang, Y., Palmese, A., et al. 2019, MNRAS, 488, 4389
Hogg, D. W. 1999, arXiv:astro-ph/9905116
Jauzac, M., Mahler, G., Edge, A. C., et al. 2019, MNRAS, 483, 3082
Jullo, E., Kneib, J. P., Limousin, M., et al. 2007, NJPh, 9, 447
Jullo, E., Natarajan, P., Kneib, J. P., et al. 2010, Sci, 329, 924
Kawai, H., & Meneghetti, M. 2024, arXiv:2410.14543
Kneib, J.-P., & Natarajan, P. 2011, A&ARv, 19, 47
Korytov, D., Hearin, A., Kovacs, E., et al. 2019, ApJS, 245, 26
Li, N., Gladders, M. D., Rangel, E. M., et al. 2016, ApJ, 828, 54
Lowing, B., Wang, W., Cooper, A., et al. 2015, MNRAS, 446, 2274
Magaña, J., Acebrón, A., Motta, V., et al. 2018, ApJ, 865, 122
Maraston, C. 2005, MNRAS, 362, 799
Mármol-Queraltó, E., McLure, R. J., Cullen, F., et al. 2016, MNRAS, 460, 3587
Massey, R., Williams, L., Smit, R., et al. 2015, MNRAS, 449, 3393
Meneghetti, M. 2022, Introduction to Gravitational Lensing: With Python Examples (Springer)
Meneghetti, M., Davoli, G., Bergamini, P., et al. 2020, Sci, 369, 1347
Meneghetti, M., Cui, W., Rasia, E., et al. 2023, A&A, 678, L2
Millon, M., Galan, A., Courbin, F., et al. 2020, A&A, 639, A101
Montes, M. 2019, arXiv:1912.01616
Montes, M., & Trujillo, I. 2014, ApJ, 794, 137
Mowla, L., Iyer, K., Asada, Y., et al. 2024, Nature, 636, 332M
Navarro, J. F., Frenk, C. S., & White, S. D. M. 1997, ApJ, 490, 493
Niemiec, A., Jauzac, M., Eckert, D., et al. 2023, MNRAS, 524, 2883
Pascale, M., Frye, B. L., Pierel, J. D. R., et al. 2024, arXiv:2403.18902
Patel, N. R., Jauzac, M., Niemiec, A., et al. 2024, MNRAS, 533, 4500
Pillepich, A., Nelson, D., Hernquist, L., et al. 2018, MNRAS, 475, 648
Purcell, C. W., Bullock, J. S., & Zentner, A. R. 2007, ApJ, 666, 20
Quinn Finney, E., Bradač, M., Huang, K.-H., et al. 2018, ApJ, 859, 1
Rangel, E., Frontiere, N., Habib, S., et al. 2020, arXiv:2008.08519
Rodriguez-Gomez, V., Pillepich, A., Sales, L. V., et al. 2016, MNRAS, 458, 2371
Strait, V., Bradač, M., Hoag, A., et al. 2018, ApJ, 868, 129
Suyu, S. H., Bonvin, V., Courbin, F., et al. 2017, MNRAS, 468, 2590
Wang, W., Li, X., Shi, J., et al. 2021, ApJ, 919, 25
Welch, B., Coe, D., Zitrin, A., et al. 2023, ApJ, 943, 2
Xie, Y., Shan, H., Li, N., et al. 2024, MNRAS, 531, 1179
Yan, Z., Shan, H., Zheng, Z., et al. 2026, RAA, 26, 024006
Zhan, H. 2011, SSPMA, 41, 1441
Zhan, H. 2021, ChSBu, 66, 1290
Zhang, Y., Yanny, B., Palmese, A., et al. 2019, ApJ, 874, 165
Zibetti, S., White, S. D. M., Schneider, D. P., & Brinkmann, J. 2005, MNRAS, 358, 949
Zwicky, F. 1937, ApJ, 86, 217
Zwicky, F. 1951, PASP, 63, 61
Zwicky, F. 1957, Morphological Astronomy